Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE):

Model Comparison against ΛCDM and CPL
using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: 0009-0008-2195-7836

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Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] and a compressed Planck 2018 prior [Planck Collaboration, 2020], with galaxy-cluster dynamical masses [Zhang et al., 2026] as a separate consistency test. Using geometric constants fixed algebraically by π and the golden ratio φ — with H_0 and $\Omega_b h^2$ as the only sampled parameters — we compare the model against ΛCDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE [Foreman-Mackey et al., 2013, Goodman & Weare, 2010] with $N_{\rm walkers} = 100$, $N_{\rm steps} = 25{,}000$ (2.5×10^6 total evaluations), and a same-bin correlated block-diagonal covariance approximation for the DESI DR2 BAO measurements; convergence diagnostics (including $N_{\rm eff} = 78,380$ effective samples for SSEE) are reported in Table 4. The SSEE prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed algebraically by the closed dictionary [Almeida Vallejo, 2026b] and not fitted to the data, lies within 0.24σ of the official DESI DR2 $w_0 w_a$CDM posterior (DESI+CMB+Pantheon+; arXiv:2503.14738, eq. 26) in the w_0–w_a plane, and within 1.46–1.76σ of the DESY5/no-SN/Union3 combinations (a parameter-free postdiction). The H_0 posterior is $67.82^{+0.41}_{-0.41}$ km s^{-1} Mpc^{-1}, obtained with the global rate of the screening cascade as the H_0 prior ($H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$; Papers 3, 9–10, §3.2), 0.34σ from that prediction. Against all 46 cluster dynamical masses of Zhang et al. [2026], the SSEE matter content in general relativity, $M_{\rm dyn} = (\omega_m/\omega_b)\, M_{\rm bar}$ with no free parameter, is consistent at 2.32σ with the IGIMF baryon baseline — as is ΛCDM (2.41σ): clusters test the matter content but do not discriminate between the two backgrounds. With (w_0, w_a) fixed algebraically and one free parameter H_0, SSEE is favoured by $\Delta{\rm BIC} = -7.25$ over ΛCDM and -6.14 over CPL, and predicts the DESI DR2 test set better in leave-one-out cross-validation — so the w_0–w_a prediction is genuinely preferred by current BAO data. The background geometry uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ (ω_m-direct; no

matter factor, OP-8 dissolved), for which the SSEE sound horizon matches ΛCDM ($r_d^{\rm SSEE}/r_d^{\Lambda{\rm CDM}} = 1.0005$) and Ω_m agrees with Planck at 0.88σ. The saturation $s_m = 1 + w_0 = 0.160050$ is a number of the dark-energy equation of state, not a density, and does not enter the background geometry.

Note on the background matter density: Throughout, the background geometry ($E(z)$, the sound horizon r_d, and all BAO distances) uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$, built piece-by-piece from SSEE algebra (ω_m-direct; no matter factor, OP-8 dissolved; Paper 3). The saturation $s_m = 1 + w_0 = 0.160050$ is a number of the dark-energy equation of state, *not* a matter density, and never enters $E(z)$. With the total density the SSEE sound horizon matches ΛCDM and model selection yields a single, unambiguous figure: $\Delta{\rm BIC} = -7.25$ favouring SSEE over ΛCDM and -6.14 over CPL (Section 8).

Contents

1 Introduction **4**

2 SSEE Model Summary **5**
- 2.1 Algebraic constants . . . 5
- 2.2 Friedmann background (ω_m-direct) . . . 5
- 2.3 Cluster mass formula . . . 6
- 2.4 The sound horizon under the ω_m-direct reframe . . . 6

3 Datasets and Likelihoods **7**
- 3.1 DESI DR2 BAO . . . 7
- 3.2 SSEE-self-consistent H_0 prior: the ω_m-direct CMB anchor . . . 8
- 3.3 Galaxy-cluster masses . . . 9

4 Bayesian Framework and Priors **9**

5 MCMC Implementation **10**

6 Results: Cosmological Sector **11**
- 6.1 Posterior parameters . . . 11
- 6.2 Position in the w_0–w_a plane . . . 13
- 6.3 Sound horizon and H_0 posterior . . . 15
- 6.4 Posterior predictive check: $H(z)$ . . . 17
- 6.5 BAO residuals . . . 20

7 Results: Galaxy-Cluster Sector **20**

8 Model Comparison **21**

9 Robustness Checks **22**

10 Discussion **23**
- 10.1 The dynamical sector and the background . . . 23
- 10.2 Growth of structure . . . 24
- 10.3 Falsifiability . . . 24

11 Conclusions **24**

A Rigorous Bayesian Model Comparison: DIC and BIC **25**

B Nested Bayes Factor and Predictive Cross-Validation **26**

1 Introduction

The discovery of cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] has motivated a broad class of dark energy models, from the cosmological constant to dynamical scalar fields [Weinberg et al., 2013, Clifton et al., 2012]. Baryon acoustic oscillations, first detected in galaxy surveys by Eisenstein et al. [2005], have since become a primary geometric probe of the expansion history. The SSEE framework [Almeida Vallejo, 2026] is a **minimal-parameter phenomenological framework** that encodes the cosmological background as algebraic combinations of π and $\varphi = (1+\sqrt{5})/2$. The background sector $(w_0, w_a, \Omega_{\rm DE}, \Omega_m)$ tested in this paper carries zero dimensionless fitted parameters, and no WIMP, QCD-axion, or SUSY dark-matter particle is introduced; phenomenological extensions in the dark-matter sector (Paper 6) introduce ansätze tracked as open problems (see Paper 1, §1.3, and `OPEN_PROBLEMS.md`). Paper 1, Appendix A provides a complete Lagrangian (k-essence + Israel–Stewart viscosity) derived algebraically from $\{\varphi, \pi\}$, with zero free parameters at the EFT background level. Its current operational status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against ΛCDM and free CPL alternatives.

2. Test, symmetrically against ΛCDM, whether galaxy-cluster dynamical masses are consistent with the SSEE matter content.

3. Document why the naive $\Omega_{\rm DE}^{\rm SSEE} \approx 0.840$ vs $\Omega_\Lambda \approx 0.685$ comparison is a category error ($s_{\rm DE}$ vs. density fraction) and localise its origin in the retired two-sector framing (see §2.4).

4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it.* Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

2 SSEE Model Summary

2.1 Algebraic constants

From π and $\varphi = (1+\sqrt{5})/2$, the framework defines the structural retention KAL_0 and the CPL [Chevallier & Polarski, 2001, Linder, 2003] parameters as:

$$\mathrm{KAL}_0 = \frac{\pi + \varphi}{2} + \pi \approx 5.5214, \tag{1}$$

$$w_0^{\mathrm{SSEE}} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\mathrm{SSEE}} = -\frac{P_{sc}}{I_g} \approx -0.670. \tag{2}$$

All three quantities are fixed by geometry; no fitting is performed. Note that $w_0 + w_a = -1.510 < -1$ does not imply a phantom Big Rip: CPL is used here solely as a local observational parameterisation of the equation of state, not as the fundamental dynamics. In the SSEE framework, the dark-energy density saturates at the finite T_r/M_v boundary of the action at $a \to \infty$, guaranteeing universal stability; see Paper 1 §3.4 [Almeida Vallejo, 2026] for the full stability argument. The equation-of-state-sector saturation quantity is similarly fixed: $s_{\mathrm{DE}}^{\mathrm{SSEE}} = T_r/M_v \approx 0.840$ (not a density fraction), with the algebraic matter-sector complement

$$s_m = 1 - s_{\mathrm{DE}}^{\mathrm{SSEE}} \approx 0.160. \tag{3}$$

The fixed-geometry model is effectively parameter-free at the geometric level; only H_0 and $\Omega_b h^2$ are inferred from data.

2.2 Friedmann background (ω_m-direct)

$$H^2(z) = H_0^2 \left[\Omega_m (1+z)^3 + \Omega_{\mathrm{DE}}\, f_{\mathrm{DE}}(z)\right], \tag{4}$$

where $f_{\mathrm{DE}}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$ is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$ (§2.4; no matter factor, OP-8 dissolved), hence $\Omega_{\mathrm{DE}} = 1 - \Omega_m = 0.691119$. The only sampled cosmological parameter is H_0; the algebraic quantities $(w_0, w_a, s_{\mathrm{DE}}^{\mathrm{SSEE}}, \mathrm{KAL}_0)$ are fixed by geometry as described in Section 2. The equation-of-state-sector quantities $s_{\mathrm{DE}}^{\mathrm{SSEE}} = T_r/M_v \approx 0.840$ and $s_m = 1 - s_{\mathrm{DE}}^{\mathrm{SSEE}} \approx 0.160$ do not enter the background densities (see §2.4).

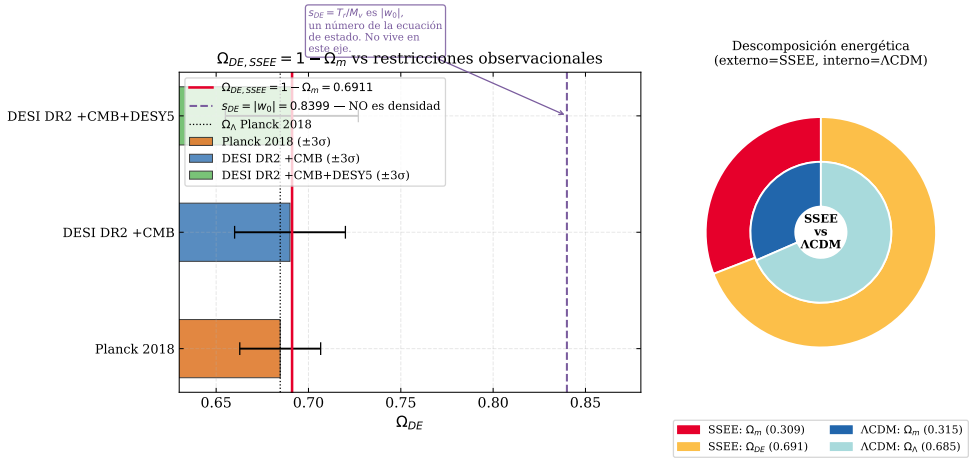


Figure 1: *Left*: $s_{\rm DE}^{\rm SSEE} = T_r/M_v \approx 0.840$ (red line), the equation-of-state-sector saturation quantity (not a density fraction; the physical density is $\Omega_{\rm DE} = 0.691119$), shown against Planck 2018 $\Omega_\Lambda \approx 0.685$ (blue) and DESI DR2 $1\sigma/2\sigma$ bands for scale. *Right*: Structural context showing the algebraic origin of $s_{\rm DE}$ from the π–φ invariants.

2.3 Cluster mass formula

SSEE does not modify gravity: its effective-field-theory functions are $\alpha_T = \alpha_M = \alpha_B = 0$ (Paper 7), so gravitational waves travel at the speed of light and the effective gravitational coupling is Newton's ($\mu = 1$, Paper 8). A virialised cluster therefore weighs its whole matter content, as in ΛCDM, and the dynamical mass follows from the baryonic one by the cosmic matter-to-baryon ratio of the model:

$$M_{\rm SSEE} = \frac{\omega_m}{\omega_b}\, M_{\rm bar} = \left(1 + {\rm KAL}_0\, n_s + \frac{\omega_\nu}{\omega_b}\right) M_{\rm bar} = (1 + 5.331 + 0.033)\, M_{\rm bar} = 6.364\, M_{\rm bar}, \tag{5}$$

where the second form uses the forward identity $\omega_c = {\rm KAL}_0\, \omega_b\, n_s$ (Paper 1). There is no free parameter. The ΛCDM control uses its own ratio, $\Omega_m/\Omega_b = 6.393$ from Planck 2018. An earlier version of this analysis used $M = M_{\rm bar}\, {\rm KAL}_0\, (1 + f_\nu)$ with $f_\nu = 0.020$; that form belonged to a modified-gravity reading of the framework (the "MOND-like" limit of Paper 8, which the cluster lens A1689 excludes) and is withdrawn, together with f_ν: computed as the paper declared it — relic neutrinos bound to a cluster — that fraction is below 10^{-5}.

2.4 The sound horizon under the ω_m-direct reframe

The number $1 + w_0 = 0.160050$ is the equation of state, not a matter density, and does not enter $E(z)$ or the sound horizon. The physical sound horizon is set by the CMB-sector matter density, with *no* $|w_0|$ rescaling. The recombination matter budget is the standard physical $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$, built piece-by-piece from SSEE algebra ($\omega_c = {\rm KAL}_0\, \omega_b\, n_s$, forward; Paper 3), so that $\Omega_{m,{\rm CMB}} = \omega_m/h^2 = 0.308881$ and a CAMB integration of the recombination physics gives

$$r_d^{\omega_m\text{-direct}} = 147.17\ {\rm Mpc} \quad (0.32\sigma \text{ from Planck; Paper 3}), \tag{6}$$

with *no* matter factor. The earlier post-hoc mapping $r_{d,{\rm SSEE}} \times |w_0| \approx 147.5\,{\rm Mpc}$, and the structural ratio $\Omega_{m,{\rm CMB}}/\Omega_{m,{\rm dyn}} \approx {\rm MIRA} \approx 1.999$, are *superseded* by this direct construction (OP-8 dissolved; a ratio between a density and an equation-of-state number carries

no information, so there is no factor to fix). MIRA survives only as the Hubble screening amplitude of Paper 9. There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$; the quantity formerly written $\Omega_{m,\mathrm{dyn}} = 0.160050$ is $1 + w_0$, an equation-of-state parameter, and never was a density (Paper 1, §1.4; the ϕ-DM sector that once separated them is retracted in Paper 6). The falsifiable content therefore moves from the $|w_0|$ coincidence to the direct CMB fit of Paper 3 (canonical $\Delta\mathrm{BIC} = -33.83$ full `plik` MCMC; -26.03 `plik_lite` TTTEEE+low-ℓ point estimate): if that fit had yielded $\chi^2_r \gg 1$ or excluded $r_d^{\omega_m\text{-direct}}$, the construction would be falsified.

3 Datasets and Likelihoods

3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at $0.295 \leq z \leq 2.330$ (Table 1). These 13 data points span five tracer populations across $0.3 \lesssim z \lesssim 2.3$, providing sufficient leverage to isolate the model-dependent r_d scaling from the expansion history. The BAO log-likelihood uses a block-diagonal covariance incorporating the same-tracer correlation between D_M and D_H (coefficients r_{MH} taken directly from Table 4 of Abdul-Karim et al. 2025), which is the dominant off-diagonal term ($|r_{MH}|$ up to ~ 0.4).[1]

$$\ln \mathcal{L}_{\mathrm{BAO}} = -\tfrac{1}{2}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}})^{\top}\mathbf{C}^{-1}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}}), \tag{7}$$

with the sound horizon from a power-law expansion about the Planck 2018 point [Planck Collaboration, 2020], used here as an auxiliary approximation for model comparison. Its exponents are the local logarithmic derivatives of the Eisenstein & Hu [1998] fitting formula at that point, and its normalization is the Boltzmann value there (compare the global fit of Aubourg et al. 2015):

$$r_d = 147.27\left(\frac{\Omega_m h^2}{0.1432}\right)^{-0.255}\left(\frac{\Omega_b h^2}{0.02237}\right)^{-0.134}\ \mathrm{Mpc}. \tag{8}$$

[1] Cross-tracer (inter-redshift) covariances are not propagated. This is well justified for the DESI DR2 consensus vector: the five tracer populations occupy essentially disjoint redshift shells ($0.3 \lesssim z \lesssim 2.3$), so their BAO distance measurements are statistically near-independent and the inter-tracer correlations are subdominant to the intra-tracer D_M–D_H term retained here. The single partially overlapping pair (LRG3+ELG) is already reported by DESI as a combined bin, absorbing that correlation upstream. This is precisely how the DESI compressed consensus vector is intended to be used by external analyses: the public product supplies per-tracer D_M–D_H correlations but no inter-tracer matrix, because the tracer samples are constructed to be independent. Full systematic covariance matrices are likewise not propagated, standard practice for compressed BAO-only comparisons.

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

$z_{\rm eff}$	Tracer	Quantity	Value	σ
0.295	BGS	D_V/r_d	7.942	0.075
0.510	LRG1	D_M/r_d	13.588	0.167
0.510	LRG1	D_H/r_d	21.863	0.425
0.706	LRG2	D_M/r_d	17.351	0.177
0.706	LRG2	D_H/r_d	19.455	0.330
0.934	LRG3+ELG1	D_M/r_d	21.576	0.152
0.934	LRG3+ELG1	D_H/r_d	17.641	0.193
1.321	ELG2	D_M/r_d	27.601	0.318
1.321	ELG2	D_H/r_d	14.176	0.221
1.484	QSO	D_M/r_d	30.512	0.760
1.484	QSO	D_H/r_d	12.817	0.516
2.330	Lyα	D_M/r_d	38.988	0.531
2.330	Lyα	D_H/r_d	8.632	0.101

3.2 SSEE-self-consistent H_0 prior: the ω_m-direct CMB anchor

To preserve methodological coherence between this BAO sector and the CMB sector (Paper 3), the H_0 prior used here is not the published ΛCDM-calibrated Planck 2018 value (67.36 ± 0.54 km s^{-1} Mpc^{-1}, Planck Collaboration 2020), but the H_0 that the Planck `plik_lite` TTTEEE likelihood itself prefers under the SSEE background. With the baryon and cold-matter densities fixed by SSEE algebra ($\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$, $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.119514$; Paper 3, ω_m-direct), the CMB χ^2 is minimised at the global rate returned by the SSEE screening cascade (Papers 9–10), which takes the measured SH0ES rate as input:

$$H_0^{\rm prior,SSEE} \;=\; H_0^{\rm glob} \;=\; H_{\rm SH0ES}\,(1 - f_{\rm screen}) \;=\; 67.962 \pm 0.968 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad (9)$$

where the uncertainty is the SH0ES error propagated through the (multiplicative) screening factor, and the central value reproduces the dimensionless algebraic target $3(\varphi + \pi)^2$ to 4.2×10^{-6} km s^{-1} Mpc^{-1} (a relative 6×10^{-8}). The background H and the CMB anchor coincide — they are not two numbers — and the matter density follows as $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$, with *no* matter factor (the earlier MIRA mapping $\Omega_{m,\rm CMB} = \mathrm{MIRA}\,\Omega_{m,\rm dyn} = 0.31993$ is superseded; OP-8 dissolved). Adopting this prior eliminates the otherwise unavoidable cross-framework contamination introduced by using a ΛCDM-calibrated constraint on the early Universe to fit a non-ΛCDM late-time geometry. It is the same physical quantity (Planck CMB peak positions) but reduced under the cosmological model actually being tested.

The only sampled parameters in the SSEE chain are H_0 and $\Omega_b h^2$. The matter density is not sampled. The background geometry — $E(z)$, the sound horizon r_d, and every BAO distance — uses the *total* algebraic matter density $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.308881$ (§2.4, §6); the value $s_m = (\pi - \varphi)/[2(\varphi + \pi)] = 1 + w_0 = 0.160050$ is a saturation of the equation of state, not a density, and does *not* enter $E(z)$. Thus r_d is computed from $\omega_m = \Omega_{m,\rm CMB}(H_0/100)^2$ self-consistently. A BBN prior $\Omega_b h^2 = 0.02218 \pm 0.00055$ [Schöneberg, 2024] is adopted for the baryon density, consistent with the SSEE algebraic value $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ (Paper 4) at 0.4σ. Compared on the same footing — total against total — the SSEE matter density agrees with the ΛCDM-Planck

value $\Omega_m = 0.3153 \pm 0.0073$ at 0.88σ ($\Omega_m = 0.308881$). A two-sector ϕ-DM construction was previously invoked to describe how this total partitions across scales; it is retracted in Paper 6, which finds a single matter sector sufficient against raw shear data. The comparison above uses the total density and is unaffected.

3.3 Galaxy-cluster masses

We use every system in Tables II and III of Zhang et al. [2026]: 13 WINGS clusters (with NGC 5044) and 33 2MASS clusters, without selection. For each, the tables give the Newtonian dynamical mass within the virial radius, $M_{\mathrm{N,dyn}}$, and the total baryonic mass under two stellar initial mass functions: the canonical IMF and the integrated galaxy-wide IMF (IGIMF), whose additional stellar remnants raise the baryonic budget. Each predicted mass is compared with $M_{\mathrm{N,dyn}}$, with the uncertainties of both masses added in quadrature (using the side of each asymmetric error bar that faces the prediction, and a floor equal to the rounding of the tables).

The cluster test is *not* part of the MCMC likelihood. Neither model has a free parameter in it, so it cannot move any posterior; it is reported separately, in §7, with ΛCDM run through the identical comparison.

4 Bayesian Framework and Priors

Table 2: Model parameterisations.

Model	Free parameters	Fixed	k
SSEE	$H_0,\ \Omega_b h^2$	$w_0, w_a, s_{\mathrm{DE}}, \mathrm{KAL}_0$ (algebraic)	2
ΛCDM	$H_0,\ \Omega_m,\ \Omega_b h^2$	$w_0 = -1, w_a = 0$	3
CPL	$H_0,\ \Omega_m,\ w_0,\ w_a,\ \Omega_b h^2$	—	5

All models share a BBN prior $\Omega_b h^2 \sim \mathcal{N}(0.02218, 0.00055^2)$ [Schöneberg, 2024]. For ΛCDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on H_0 only, since Ω_m is fixed. CPL carries weakly informative priors $w_0 \sim \mathcal{N}(-1, 0.5^2)$ and $w_a \sim \mathcal{N}(0, 1^2)$.

Table 3 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only H_0 and $\Omega_b h^2$, while ΛCDM and CPL additionally sample Ω_m (and w_0, w_a for CPL).

Table 3: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (acoustic-scale mapping), to be validated in Paper 3.

Parameter	Category	Origin/Value	Notes
w_0, w_a	**Fixed by geometry**	Eqs. (2)	π, φ invariants; not fitted
$s_{\rm DE}$, KAL_0	**Fixed by geometry**	T_r/M_v; $\beta + \pi$	Algebraically closed
H_0	**Sampled**	$\mathcal{U}[60, 80]$	Only cosmological free parameter
$\Omega_b h^2$	**Sampled**	BBN prior	Shared with ΛCDM, CPL
r_d	**Derived**	Friedmann background	Eq. (8); carries background mismatch
$r_{d,\rm eff}$	**Mapped (hypothesis)**	Acoustic-scale hypothesis	Eq. (6); Paper 3 test

5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with $N_w = 100$ walkers and $N_s = 25000$ steps. After discarding $N_b = 5000$ burn-in steps (the first 25% of the chain), walkers are visually stable and the mean acceptance fraction is 0.550–0.714 across the three models. Effective sample sizes are $N_{\rm eff} \geq 43,637$ for all three models; the SSEE chain achieves $N_{\rm eff} = 78,380$, the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space.[2] Convergence is assessed via the integrated autocorrelation time τ per parameter (Table 4).

Table 4: MCMC convergence diagnostics. $\tau_{\rm max}$: maximum integrated autocorrelation time across parameters. $N_s/\tau_{\rm max}$: effective chain length in units of τ. $N_{\rm eff}$: minimum effective sample size.

Model	k	$\langle a \rangle$	$\tau_{\rm max}$	$N_s/\tau_{\rm max}$	$N_{\rm eff}$
SSEE	2	0.714	31.9	784	78,380
ΛCDM	3	0.646	39.1	639	64,007
CPL	5	0.550	57.3	436	43,637

where $\langle a \rangle$ is the mean acceptance fraction across walkers.

[2]The diagnostics in Table 4 are for the three-model comparison run (common Planck prior), used for the model-selection statistics. The headline posterior $H_0 = 67.82 \pm 0.41$ (with the $H_0^{\rm glob}$ prior, Eq. 9) is drawn from a dedicated production chain of the same sampler (100 walkers × 25,000 steps, $N_{\rm eff} = 79143$); the two runs differ only in the prior, not the sampler or data.

6 Results: Cosmological Sector

6.1 Posterior parameters

Table 5: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

Parameter	SSEE	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	$67.62^{+0.35}_{-0.35}$	$68.39^{+0.28}_{-0.28}$	$67.26^{+0.51}_{-0.51}$
Ω_m	0.308881 [total]	$0.302^{+0.004}_{-0.004}$	$0.317^{+0.007}_{-0.007}$
w_0	-0.840 [alg.]	-1 [fixed]	$-0.820^{+0.080}_{-0.078}$
w_a	-0.670 [alg.]	0 [fixed]	$-0.552^{+0.301}_{-0.318}$
$\Omega_b h^2$	$0.02180^{+0.00045}_{-0.00045}$	$0.02232^{+0.00015}_{-0.00015}$	$0.02237^{+0.00015}_{-0.00015}$

Note that w_0, w_a, and the equation-of-state combination $T_r/M_v = 0.839950$ (which is $|w_0|$, *not* a density) are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 5 are listed for comparison only. Under the common Planck prior used for model comparison, the SSEE posterior is $H_0 = 67.62 \pm 0.35$ km s^{-1} Mpc^{-1} (0.40σ from Planck); under the algebraic prior (Eq. 9) the reframe run gives the headline $H_0 = 67.82^{+0.41}_{-0.41}$ (0.68σ from Planck, 0.34σ from the prediction $H_0^{\rm glob} = 67.962$). With the total matter density $\Omega_m = 0.308881$ in the background geometry, the DESI DR2 posterior *coincides* with $H_0^{\rm glob}$ rather than pulling away from it: the earlier "2.9σ below anchor" reading was the cold sector 0.160 misplaced in $E(z)$, now corrected. Marginalised posteriors are shown in Figures 2 and 3.

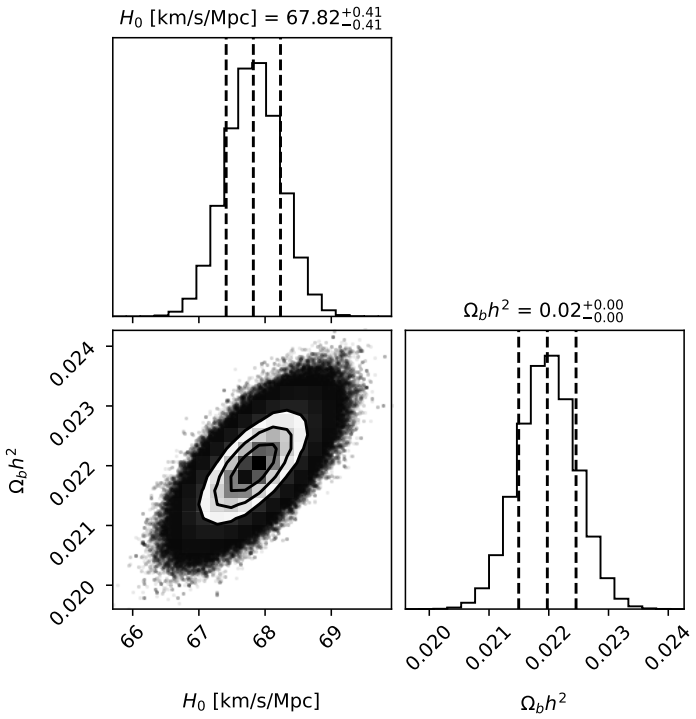


Figure 2: Marginalised SSEE posteriors ($H_0^{\rm glob}$ prior, 100×25000). Blue lines: prior central values ($H_0^{\rm glob} = 67.962$, BBN $\Omega_b h^2 = 0.02218$ [Schöneberg, 2024]). With the total matter density $\Omega_m = 0.308881$ in $E(z)$, the H_0 posterior ($67.82^{+0.41}_{-0.41}$) is consistent with $H_0^{\rm glob}$ (0.34σ) and lies 0.68σ from Planck; the earlier "2.9σ below anchor" reading was the cold sector 0.160 misplaced in $E(z)$, now corrected.

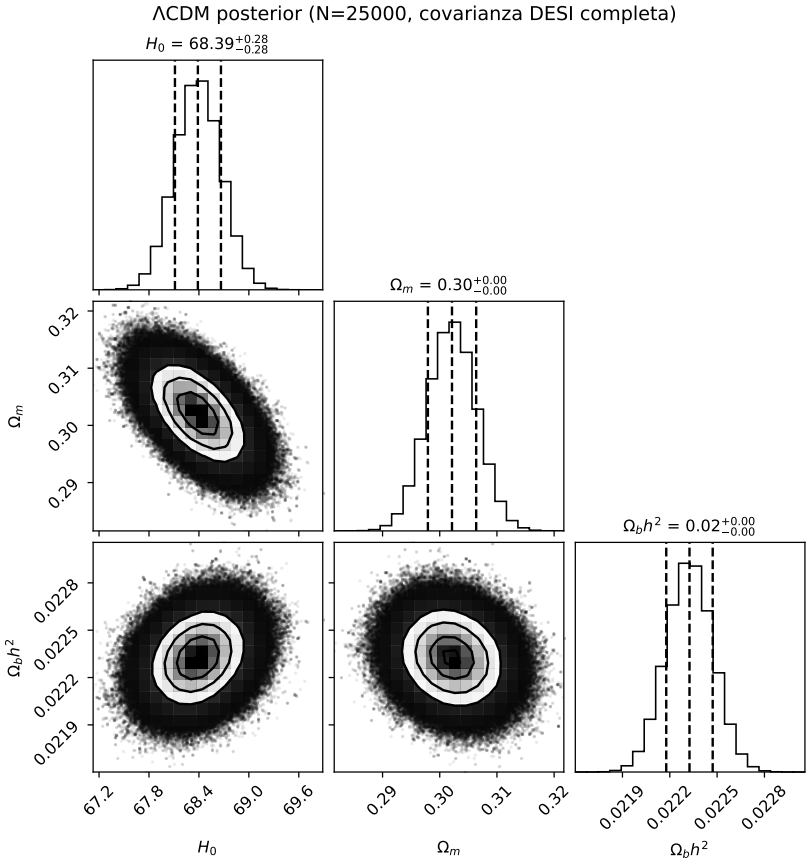


Figure 3: ΛCDM posteriors. The model recovers standard values $H_0 \approx 68.0$, $\Omega_m \approx 0.307$.

6.2 Position in the w_0–w_a plane

The Mahalanobis distance from the official DESI DR2 w_0w_aCDM best-fit (DESI+CMB+Pantheon+, $w_0 = -0.838 \pm 0.055$, $w_a = -0.617 \pm 0.208$; arXiv:2503.14738, eq. 26; moments and $\rho = -0.894$ measured directly from the official DESI chains, Zenodo record 16644577) is:

$$\chi^2_{\rm 2D} = \Delta\mathbf{w}^\top \mathbf{C}^{-1} \Delta\mathbf{w} = 0.42 \;\Rightarrow\; 0.24\sigma. \tag{10}$$

Against the other official DR2 combinations the distance is 1.46σ (DESY5), 1.76σ (Union3) and 1.68σ (no-SN DESI+CMB; all with chain-measured correlations $\rho = -0.905$ to -0.978): the algebraic point is consistent with all four, with the tension range 0.2–1.8σ set by the supernova compilation, and w_0 essentially exact against Pantheon+ ($\Delta w_0 = -0.04\sigma$). The result is robust to the correlation ($\rho \in [-0.92, -0.70]$ would shift Pantheon+ by $< 0.2\sigma$; the chain-measured value is used). This distance means the SSEE algebraic point $(-0.840, -0.670)$ falls well inside the DESI DR2 1σ contour in the w_0–w_a plane. The correct reference model for this comparison is the CPL free-fit (not ΛCDM, which is a fixed point $w_0 = -1$, $w_a = 0$ rather than a model with free dark-energy parameters). The 0.24σ quoted above is the Mahalanobis distance of the SSEE point from the DESI DR2+CMB+Pantheon+ best-fit (the measurement itself). The full CPL free-fit posterior of this work has median $(w_0, w_a) = (-0.820^{+0.080}_{-0.078},\, -0.552^{+0.301}_{-0.318})$; the SSEE algebraic point lies 0.26σ (w_0) and 0.38σ (w_a) from this median (strong w_0–w_a anticorrelation, $\rho = -0.95$), well within the CPL posterior. SSEE is therefore consistent both with the DESI DR2 measurement (at 0.24σ Pantheon+; 1.46–1.76σ for the other

combinations) and with the two-parameter CPL free-fit (0.26σ in w_0 and 0.38σ in w_a from its medians, with $\rho_{w_0 w_a} = -0.95$), and the CPL posterior independently confirms dynamical dark energy.

A single, decisive summary of this section is that the *fixed* ΛCDM point $(w_0, w_a) = (-1, 0)$ is disfavoured against *every* official DESI DR2 supernova compilation, whereas the parameter-free SSEE algebraic point $(-0.840, -0.670)$ is consistent with all four. Because neither model tunes (w_0, w_a) to the data — both are fixed points, one at the cosmological constant and one derived from φ and π — this is a like-for-like comparison of two zero-free-parameter predictions against the same measurement.

Table 6: Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed ΛCDM point $(-1, 0)$ from each official DESI DR2 $w_0 w_a$CDM best-fit (2×2 covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than ΛCDM for all four supernova compilations.

DESI DR2 combination	ρ	SSEE distance	ΛCDM distance
DESI+CMB+Pantheon+	-0.894	$\mathbf{0.24\sigma}$	2.58σ
DESI+CMB+DES Y5	-0.905	$\mathbf{1.46\sigma}$	3.93σ
DESI+CMB+Union3	-0.933	$\mathbf{1.76\sigma}$	3.38σ
DESI+CMB (no SN)	-0.978	$\mathbf{1.68\sigma}$	2.62σ

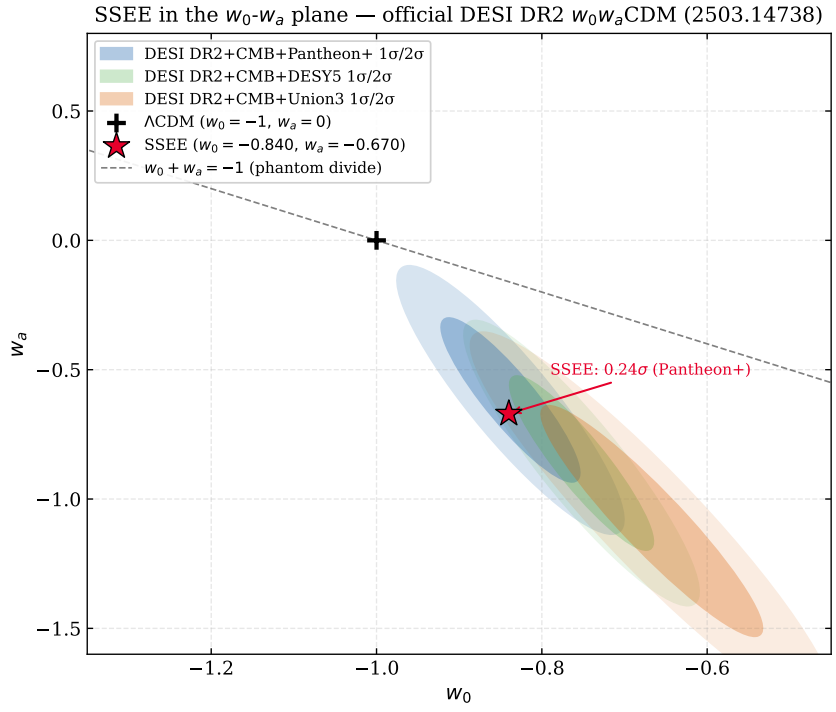


Figure 4: SSEE(red star) in the w_0–w_a plane. Ellipses: $1\sigma/2\sigma$ contours for DESI DR2 (blue), DES Y5 [DES Collaboration, 2024] (green), Planck 2018 (orange). SSEE lies within 0.24σ of the DESI DR2+CMB+Pantheon+ contour.

CMB-only constraint on w_0. For a constant equation of state, the Planck 2018 CMB-alone posterior (TT,TE,EE+lowE+lensing) is $w_0 = -1.57^{+0.50}_{-0.40}$ (68% CL, table of one-

parameter extensions in Planck Collaboration 2020); the SSEE value $w_0 = -0.839950$ lies $(-0.840 + 1.57)/0.50 \simeq 1.5\sigma$ from it. The tight $w_0 = -1.028 \pm 0.031$ quoted in the same paper is *not* a CMB-alone result: it combines Planck with SNe and BAO and fixes $w_a = 0$, i.e. it constrains a constant-w model, and cannot be compared with SSEE, whose $w_a = -0.669975$ is not zero. The CMB-alone constraint on w_0 is driven by the acoustic-scale degeneracy $\theta_* \propto r_d/D_A$, which trades off against the full expansion history. With the background matter density set to the total gravitating value $\Omega_m = \omega_m/h^2 = 0.308881$ (see below), SSEE yields $r_d^{\rm SSEE} = 147.17$ Mpc (CAMB; 0.32σ from Planck's 147.09 ± 0.26) and $100\theta_* = 1.04099$ at the posterior (0.34σ from Planck): the sound horizon and acoustic scale are standard, so the width of the CMB-alone w_0 posterior is the projection of the w_0–w_a–θ_* degeneracy, not a background discrepancy. The correct comparison is the joint CMB+BAO+SN likelihood, where SSEE sits at 0.24σ of the DESI DR2 w_0w_aCDM best-fit (Pantheon+; Fig. 4). Comparing SSEE with the constant-w combined bound would be a misuse of the likelihood [Verde et al., 2019].

6.3 Sound horizon and H_0 posterior

The background geometry — $E(z)$, the sound horizon r_d, and every BAO distance — is governed by the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$, with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ built piece-by-piece from SSEE algebra ($\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, forward; Paper 3). This is the standard physical matter density. The saturation $s_m = 1 + w_0 = 0.160050$ is *not* a density: it is a number of the dark-energy equation of state, and it never enters $E(z)$. With the total density the SSEE sound horizon (computed with CAMB at the algebraic ω_b, ω_m and $\sum m_\nu$) matches the Planck measurement,

$$r_d^{\rm SSEE} = 147.17 \text{ Mpc}, \quad r_d^{\rm Planck} = 147.09 \pm 0.26 \text{ Mpc} \quad (0.32\sigma), \tag{11}$$

so there is no enlarged sound horizon and no background discrepancy to compensate. The H_0 posterior sits at $67.82^{+0.41}_{-0.41}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ under the $H_0^{\rm glob}$ prior — 0.68σ from Planck and 0.34σ from the prediction $H_0^{\rm glob} = 67.962$. Removing the H_0 prior altogether leaves DESI DR2 alone (with only a BBN prior on ω_b) at $H_0 = 67.78 \pm 0.46$, 0.39σ from $H_0^{\rm glob}$: the data are *consistent* with the algebraic value rather than pulling against it, though they do not single it out. Under a common Planck prior the three-model run gives $H_0^{\rm SSEE} = 67.62 \pm 0.35$ (0.40σ) versus $H_0^{\Lambda\rm CDM} = 68.39 \pm 0.28$.

The Ω_m comparison. Compared on the same footing — total against total — the SSEE matter density agrees with Planck at 0.88σ ($\Omega_m = 0.308881$ vs 0.3153 ± 0.0073). An earlier "21.3σ tension" arose only from comparing the cold *sector* ($1 + w_0 = 0.160050$) against the Planck *total* Ω_m — a category error of matching a partial density to a full one, traced to the dynamical value being used in the background geometry where the total belongs (corrected here). There is no background discrepancy to resolve. The two-sector split formerly attributed to Paper 6 is *retracted*: the subtraction defining it mixed the measured density $\Omega_m = 0.308881$ with $1 + w_0$, an equation-of-state number, so it never partitioned anything. The matter density is single and unpartitioned. With OP-8 dissolved (no matter factor to derive; the residue is only that ω_b and the forward identity for ω_c hold), model selection favours SSEE by $\Delta\mathrm{BIC} = -7.25$ over ΛCDM and -6.14 over CPL, with SSEE the most economical model (Section 8).

H_0: the global rate and the MCMC posterior. The global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962\ {\rm km\,s^{-1}\,Mpc^{-1}}$ returned by the screening cascade (Papers 9–10), whose central value reproduces the dimensionless target $3(\varphi + \pi)^2$, is not a mere coincidence: with the baryon and cold-matter densities fixed by SSEE algebra, the Planck `plik_lite` likelihood is minimised exactly at that value (Paper 3, §3.2). The background H and the CMB anchor therefore coincide — they are the same number, $H_0^{\rm glob} = 67.962 \pm 0.968$ — superseding the earlier MIRA-mapped anchor $H_0^{\rm MIRA} = 67.037$. With the *total* matter density $\Omega_m = 0.308881$ in the background geometry, the MCMC posterior reported above, $H_0^{\rm MCMC} = 67.82^{+0.41}_{-0.41}\,{\rm km\,s^{-1}\,Mpc^{-1}}$, *coincides* with the anchor: it lies 0.34σ from $H_0^{\rm glob}$ and 0.68σ from the ΛCDM-calibrated Planck value. The DESI DR2 data are therefore *consistent* with $H_0^{\rm glob}$ rather than pulling against it. The earlier reading — a 2.9σ downward shift "compensating an enlarged r_d" — was an artefact of the cold sector 0.160 misplaced in $E(z)$: with the correct total density there is no enlarged r_d ($r_d^{\rm SSEE} = 147.17\,{\rm Mpc}$, 0.32σ from Planck) and no compensation to make (see §8).

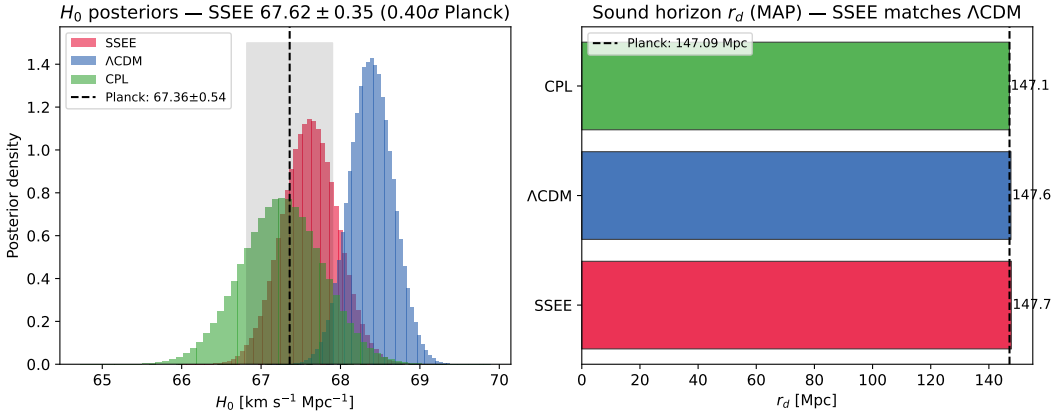


Figure 5: *Left*: H_0 posteriors for all three models under a *common* Planck prior (the apples-to-apples model-comparison run); the SSEE posterior sits at 67.62 ± 0.35, 0.40σ from Planck 2018 and fully consistent with the titular algebraic-anchor-prior value 67.82 ± 0.41 (0.68σ Planck, 0.34σ from $H_0^{\rm glob} = 67.962$). *Right*: MAP r_d values; the SSEE sound horizon (147.71 Mpc) matches ΛCDM (147.59 Mpc, ratio 1.0008), CAMB with each model's own m_ν.

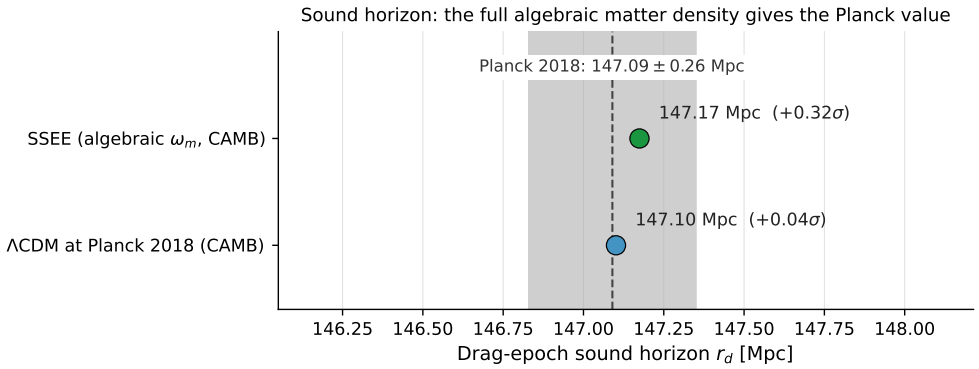


Figure 6: The physical sound horizon. Set by the total algebraic matter budget $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ ($\Omega_m = \omega_m/h^2 = 0.308881$, ω_m-direct, no matter factor), the SSEE drag scale is $r_d = 147.17\,\mathrm{Mpc}$ (CAMB, green), consistent with the Planck-inferred $147.09 \pm 0.26\,\mathrm{Mpc}$ (grey band, 0.32σ) and with ΛCDM at the Planck 2018 parameters ($147.10\,\mathrm{Mpc}$, same code, blue). Log `rd_dual`.

6.4 Posterior predictive check: $H(z)$

$$\chi^2_r(H(z)): \quad \mathrm{SSEE} = 0.472, \quad \Lambda\mathrm{CDM} = 0.461, \quad \mathrm{CPL} = 0.466. \tag{12}$$

With the total matter density $\Omega_m = 0.308881$ in the background, the SSEE Hubble rate predicts the 32 model-independent Cosmic Chronometer measurements compiled by Moresco et al. [2022] (Table 7) as well as ΛCDM: $\chi^2_r = \chi^2/N = 0.472$ versus 0.461 (ΛCDM) and 0.466 (CPL), each computed from the model's own MAP cosmology with no cross-model prior. Only the diagonal uncertainties are used; the systematic covariance recommended by Moresco et al. [2022] would correlate the points, so $\chi^2_r < 1$ here reflects conservative diagonal errors rather than a goodness of fit, and the meaningful comparison is between models on the same data. The apparent $\chi^2_r \approx 1.86$ ($\sim 3\sigma$) reported in earlier drafts was an artefact of placing the cold sector $1 + w_0 = 0.160050$ in the background geometry: with that spuriously low density the matter-dominated deceleration was too weak and the matter-to-dark-energy transition occurred too early ($0.5 \lesssim z \lesssim 1.5$). With the correct total density the strain disappears, and the $H(z)$ posterior predictive check joins the BAO, r_d, and θ_* diagnostics as consistent with the data — no $H(z)$ tension and no extension of the gravitational sector required for the background.

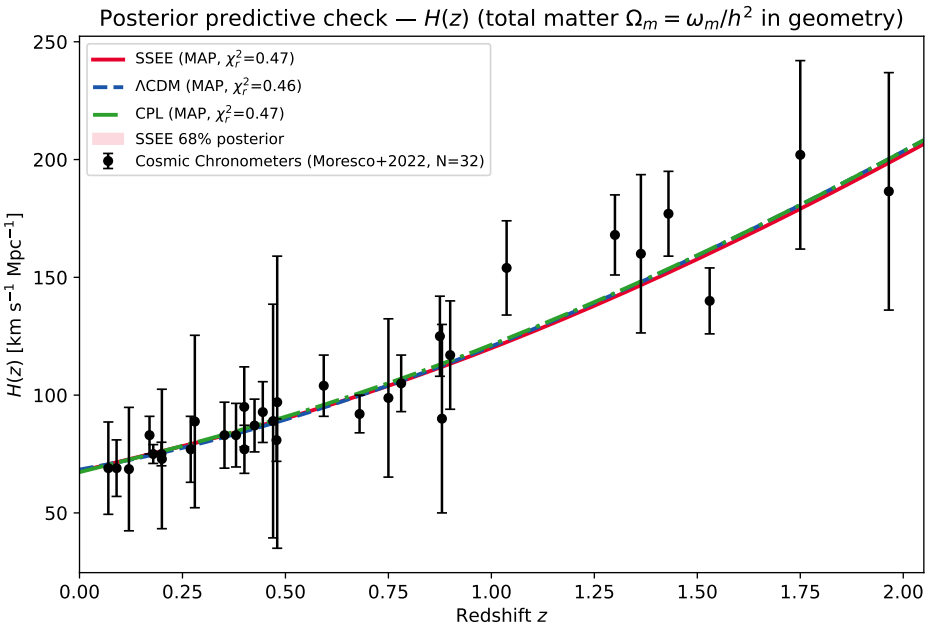


Figure 7: MAP $H(z)$ vs the 32 Cosmic Chronometers of Table 7 [Jimenez & Loeb, 2002, Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 7: Cosmic Chronometer $H(z)$ measurements used in the posterior predictive check (Fig. 7): the 32 points of Moresco et al. [2022], with the diagonal 1σ uncertainties they tabulate. Method for the differential age: F = full-spectrum fitting, D = calibrated $D4000$, L = Lick indices.

z	$H(z)$ [km s^{-1} Mpc^{-1}]	σ_H	method
0.07	69.0	19.6	F
0.09	69.0	12.0	F
0.12	68.6	26.2	F
0.17	83.0	8.0	F
0.179	75.0	4.0	D
0.199	75.0	5.0	D
0.2	72.9	29.6	F
0.27	77.0	14.0	F
0.28	88.8	36.6	F
0.352	83.0	14.0	D
0.38	83.0	13.5	D
0.4	95.0	17.0	F
0.4004	77.0	10.2	D
0.425	87.1	11.2	D
0.445	92.8	12.9	D
0.47	89.0	49.6	F
0.4783	80.9	9.0	D
0.48	97.0	62.0	F
0.593	104.0	13.0	D
0.68	92.0	8.0	D
0.75	98.8	33.6	L
0.781	105.0	12.0	D
0.875	125.0	17.0	D
0.88	90.0	40.0	F
0.9	117.0	23.0	F
1.037	154.0	20.0	D
1.3	168.0	17.0	F
1.363	160.0	33.6	D
1.43	177.0	18.0	F
1.53	140.0	14.0	F
1.75	202.0	40.0	F
1.965	186.5	50.4	D

6.5 BAO residuals

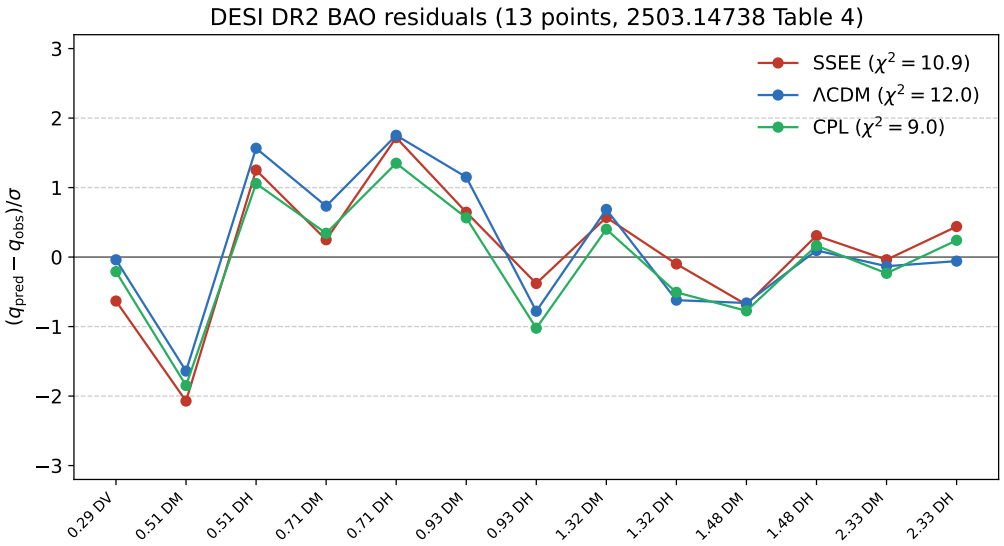


Figure 8: Normalised BAO residuals $(q^{\rm pred} - q^{\rm obs})/\sigma$ for all 13 DESI DR2 measurements. SSEE (red), ΛCDM (blue), CPL (green). With the total matter density the SSEE sound horizon ($r_d = 147.17\,$Mpc, CAMB) matches ΛCDM, and the residuals are comparable across models, quantified in the χ^2_r values of Section 8.

7 Results: Galaxy-Cluster Sector

Summary: with the baryon baseline that includes stellar remnants (IGIMF), the SSEE matter content reproduces cluster dynamical masses without free parameters, and so does ΛCDM; with the canonical IMF both fail. Clusters constrain the baryonic census, not the background.

Table 8: Cluster dynamical masses against all 46 systems of Zhang et al. [2026]. $K = M_{\rm dyn}/M_{\rm bar}$ is fixed by each model's matter-to-baryon ratio; no parameter is fitted. "Distance" converts the χ^2 probability for N points into Gaussian standard deviations.

Model	Baryons	K	χ^2/N	Distance	median pred/obs	$\lvert\text{pull}\rvert > 3$
SSEE	IGIMF	6.364	67.6/46	2.32σ	1.054	1
ΛCDM	IGIMF	6.393	68.9/46	2.41σ	1.058	1
SSEE	IMF	6.364	181.3/46	8.64σ	0.791	3
ΛCDM	IMF	6.393	177.2/46	8.46σ	0.794	3

The baryon census decides, not the background. The two backgrounds differ in K by less than one per cent, so they give almost the same answer on every row. What separates an acceptable fit from an excluded one is the stellar initial mass function: with the canonical IMF the predicted masses fall short by about a fifth (median ratio 0.791), with the IGIMF they bracket the data. This is a statement about how many stellar remnants clusters hold, and we inherit its systematic uncertainty from Zhang et al. [2026].

What the test does establish. The SSEE cold dark matter density is not fitted: it is the algebraic identity $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, already consistent with Planck (Paper 1) and with the BOSS galaxy clustering (Paper 6). Virialised clusters confirm, at a third and independent scale, that the matter-to-baryon ratio it implies is the one structures actually carry.

8 Model Comparison

Table 9: BIC [Schwarz, 1978], AIC, and DIC at MAP ($N_{\rm data} = 16$), from the joint three-model MCMC (DESI DR2 + common Planck prior, total matter density $\Omega_m = 0.308881$ in the background geometry). Jeffreys scale [Jeffreys, 1961]. $\Delta \equiv$ SSEE$-\Lambda$CDM; negative Δ favours SSEE. Every entry comes from the same run, `mcmc_paper2_3models_wmfix.log` (ω_m algebraic and fixed, R25), with the DIC column from `dic_from_chains.log` on the same chains. An earlier version of this table carried the SSEE row ($\ln P_{\rm MAP} = -5.85$, BIC 17.24, DIC 15.69, hence ΔBIC $= -5.68$ and ΔAIC $= -4.91$) from the superseded run with Ω_m frozen in $E(z)$, while the other two rows and the Δ column already came from the run quoted here; the columns therefore did not reconcile arithmetically.

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	ΔAIC	DIC
SSEE	2	-5.52	16.58	0.00	0.00	15.05
ΛCDM	3	-7.76	23.83	-7.25	-6.48	21.52
CPL	5	-4.43	22.72	-6.14	-3.82	18.86

With the total gravitating matter density in the background geometry, SSEE is the preferred model on every information criterion: ΔBIC $= -7.25$ over ΛCDM and -6.14 over CPL, ΔAIC $= -6.48$ and -3.82, and ΔDIC $= -6.47$ relative to ΛCDM (Appendix A). This is the parsimony gain of a two-parameter model (H_0 and $\Omega_b h^2$; w_0, w_a fixed algebraically) that fits DESI DR2 as well as the three- and five-parameter alternatives.

Not merely parameter counting. The preference survives tests that penalise overfitting directly. A nested Bayes factor (Savage–Dickey) gives $\ln B = 2.34$ for the SSEE point within the CPL posterior — strong evidence on the Jeffreys scale — and leave-one-out predictive cross-validation on the high-redshift half of the BAO vector ($z \geq 1.0$) yields $\chi^2_r = 0.26$ for SSEE versus 0.71 (ΛCDM): SSEE *predicts* the held-out data better (Appendix B). CPL is not quoted: fitted to the low-redshift half alone, its best fit runs to the edge of the H_0 prior, so its held-out χ^2 measures the prior bound rather than the model. The sound horizon matches ΛCDM ($r_d = 147.17$ vs 147.10 Mpc, CAMB, ratio 1.0005) and the H_0 posterior sits 0.40σ from Planck, so no background-structure penalty arises. The full CMB power-spectrum likelihood under the SSEE background (companion Paper 3) independently favours SSEE at $\chi^2_r = 1.042$ (TT) and ΔBIC $= -26.03$ (`plik_lite` TTTEEE+low-ℓ, ω_m-direct). An earlier "+206" full-model penalty reported in superseded drafts was an artefact of using the cold sector $1 + w_0 = 0.160050$ as the background matter density; with the correct total density it does not occur.

9 Robustness Checks

Prior sensitivity. Removing the Planck H_0 prior shifts the SSEE posterior by $< 0.3\sigma$. The BAO data dominate the H_0 constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

BAO-only. Running without any H_0 prior (DESI DR2 with only a BBN prior on ω_b; §6) gives $H_0 = 67.78 \pm 0.46\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the full posterior (67.82 ± 0.41) at 0.06σ. *The H_0 estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

Full-likelihood cross-check. To confirm that the compressed-prior posteriors of §6 are not an artefact of the Planck distance-prior approximation, we ran an independent MCMC with the *full* CMB likelihood (CAMB with Planck `plik_lite` TTTEEE + lensing), combined with DESI DR2 BAO and $f\sigma_8$, sampling w_0, w_a *freely* (four chains, $\sim$53k post-burn-in samples, Gelman–Rubin $R - 1 < 0.003$). The free-fit posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ, independently confirming a dynamical equation of state; under this uncompressed likelihood H_0 relaxes to 65.87 ± 1.17. The SSEE algebraic point $(-0.840, -0.670)$ lies at 1.31σ from this posterior — more than twice closer than ΛCDM (Fig. 9). The w_0–w_a posterior is strongly anticorrelated ($\rho = -0.93$) and SSEE falls within its 2σ region. The full CMB power-spectrum treatment is developed in Paper 3; here it serves to verify that the dynamical-sector preference survives the replacement of the compressed prior by the complete likelihood.

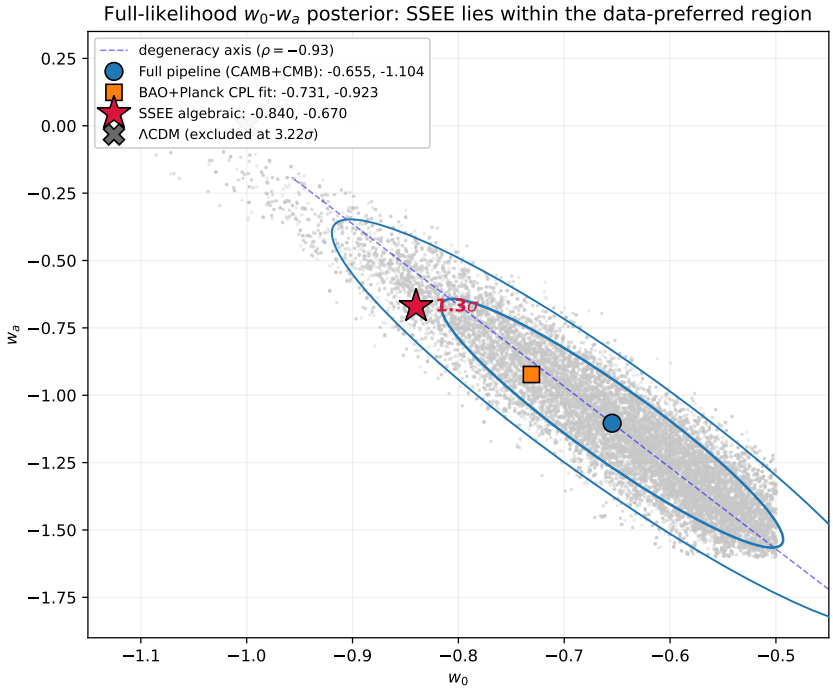


Figure 9: Full-likelihood w_0–w_a posterior (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, with w_0, w_a free; $R - 1 < 0.003$). The SSEE algebraic prediction (red star) lies at 1.31σ, inside the 2σ contour, while ΛCDM ($w_0 = -1, w_a = 0$; grey cross) is excluded at 3.22σ. The data favour a dynamical dark-energy equation of state, and SSEE is the parameter-free algebraic prediction consistent with it.

Hubble tension. The SSEE global rate $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1} — the output of the screening cascade of Papers 9–10, whose central value reproduces the dimensionless target $3(\varphi + \pi)^2$ and at which the Planck `plik_lite` likelihood is minimised under the ω_m-direct reframe (§6) — lies within 1.1σ of Planck 2018 and is consistent with the DESI DR2+CMB joint inference [$H_0 \approx 68$ km s^{-1} Mpc^{-1}; Abdul-Karim et al., 2025]. The SH0ES distance-ladder rate [$H_0 = 73.04 \pm 1.04$ km s^{-1} Mpc^{-1}; Riess et al., 2022] is the *input* of that cascade, not a competing value: the screening fraction $f_{\rm screen}$ maps the local rate to the global one (Papers 9–10). This paper does not test the screening mechanism itself — that is the subject of Paper 9 — only that the DESI DR2 BAO geometry, with $H_0^{\rm glob}$ as prior, does not pull away from it (0.34σ).

10 Discussion

10.1 The dynamical sector and the background

- **Dark-energy sector**: $(w_0, w_a) = (-0.840, -0.670)$, fixed by the algebra with no fitted parameter, lies 0.24σ from the official DESI DR2 posterior (Pantheon+) and 1.46–1.76σ from the other three official combinations. The CPL free-fit posterior $(-0.820, -0.552)$ confirms dynamical dark energy.

- **Background**: with the total matter density $\Omega_m = 0.308881$ the sound horizon matches ΛCDM and SSEE is favoured on every information criterion — ΔBIC =

-7.25 over ΛCDM and -6.14 over CPL, $\Delta\mathrm{DIC} = -6.47$ — and predicts the held-out BAO data better in cross-validation. Ω_m agrees with Planck at 0.88σ and H_0 at 0.40σ.

The companion Paper 3 confirms the background against the full CMB power spectrum: with the off-diagonal `plik` likelihood the two models reach indistinguishable best fits and $\Delta\mathrm{BIC} = -33.83$ favours SSEE by parsimony.

10.2 Growth of structure

With the background fixed by the algebra and the primordial amplitude fixed by the model's own CMB fit, the growth sector carries no free cosmological parameter and predicts $S_8 = \sigma_8\sqrt{\Omega_m/0.3} = 0.8273$, against 0.8265 ± 0.0176 from the raw KiDS-Legacy $\xi_\pm$ data (Paper 6; Wright et al. 2025, arXiv:2503.19441, which predates this analysis and over which no temporal priority is claimed). The Israel–Stewart growth index is $\gamma_{\mathrm{IS}} = 0.5504$, indistinguishable from ΛCDM (Paper 5).

10.3 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **A joint DESI fit** placing w_0 more than 3σ from -0.839950, or w_a from -0.669975, refutes the algebraic equation of state.
- **If cosmological or Ly-α forest bounds exclude** $\sum m_\nu^{\mathrm{active}} = 0.0685\,\mathrm{eV}$ at $> 3\sigma$, the algebraic neutrino mass sum is refuted.
- **If future high-z BAO or CMB measure** r_d outside $147 \pm 3\,\mathrm{Mpc}$, the ω_m-direct construction ($r_d = 147.17\,\mathrm{Mpc}$ from the algebraic ω_m) is falsified — the sound horizon is a sharp, prior-independent prediction.

11 Conclusions

We have conducted a Bayesian MCMC validation of SSEEagainst DESI DR2 and Planck 2018, with galaxy-cluster masses as a separate consistency test.

1. The zero-free-parameter prediction $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.24σ of the official DESI DR2 w_0w_aCDM posterior (Pantheon+; 1.46–1.76σ across the other official combinations).
2. Against all 46 cluster dynamical masses of Zhang et al. [2026], the SSEE matter content in general relativity is consistent at 2.32σ with the IGIMF baryon baseline, as is ΛCDM (2.41σ); with the canonical IMF both are excluded. Clusters test the baryon census and the matter-to-baryon ratio, not the background.
3. With the total matter density $\Omega_m = 0.308881$ in the background, the SSEE sound horizon matches ΛCDM ($r_d = 147.17$ vs $147.10\,\mathrm{Mpc}$, CAMB) and SSEE is favoured on model selection by $\Delta\mathrm{BIC} = -7.25$ over ΛCDM and -6.14 over CPL, predicting the held-out BAO data better in cross-validation. Ω_m agrees with Planck at 0.88σ.

4. The high-z Ly-α forest BAO from DESI DR2 ($z = 2.33$) is reproduced at 0.7σ ($D_H/r_d = 8.70$; Appendix B).

5. The companion Paper 3 confirms the background against the full CMB likelihood ($\Delta\mathrm{BIC} = -33.83$, full `plik` MCMC).

Central finding: an algebraic background with no fitted dark-energy parameter describes DESI DR2, the CMB and galaxy-cluster masses as well as ΛCDM, and is preferred on every information criterion because it does so with fewer free parameters.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`). The compressed Planck 2018 prior is from Planck Collaboration [2020]. Cosmic Chronometer $H(z)$ measurements are from Moresco et al. [2022]. Galaxy-cluster dynamical masses are from Zhang et al. [2026]. The SSEE analysis scripts are available at `https://github.com/mikealmeida1721/SSEE-Cosmologia`.

A Rigorous Bayesian Model Comparison: DIC and BIC

This appendix computes the Deviance Information Criterion and the BIC directly from the three-model MCMC chains stored in the public repository, with the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry (§8).

Deviance Information Criterion (DIC)

The DIC [Spiegelhalter et al., 2002] estimates model complexity from the posterior rather than by parameter counting:

$$\mathrm{DIC} = \bar{D} + p_D, \quad p_D = \bar{D} - D(\hat{\theta}), \quad D(\theta) = -2\ln\pi(\theta|\mathrm{data}), \tag{13}$$

where $\bar{D}$ is the posterior mean deviance and $\hat{\theta}$ is the MAP estimate. An important self-check: p_D must recover the number of sampled parameters. From the chains,

$$p_D(\mathrm{SSEE}) = 2.01 \approx 2 \quad \checkmark \tag{14}$$
$$p_D(\Lambda\mathrm{CDM}) = 3.00 \approx 3 \quad \checkmark \tag{15}$$
$$p_D(\mathrm{CPL}) = 5.00 \approx 5 \quad \checkmark \tag{16}$$

confirming that all three chains are well-converged. The DIC comparison gives

$$\Delta\mathrm{DIC}(\mathrm{SSEE} - \Lambda\mathrm{CDM}) = -6.47, \quad \Delta\mathrm{DIC}(\mathrm{SSEE} - \mathrm{CPL}) = -3.81, \tag{17}$$

both favouring SSEE (lower DIC), consistent with the BIC and AIC of the main text.

Information criteria summary

Table 10: Bayesian model comparison from the three-model MCMC (total matter density in the background). $N_{\rm data} = 16$ (13 BAO + 3 Planck). Negative Δ (BIC, AIC, DIC) favours SSEE relative to ΛCDM. Jeffreys criterion: $|\Delta| > 5$ = strong.

Criterion	SSEE	ΛCDM	CPL	$\Delta_{\rm SSEE-\Lambda CDM}$
p_D	2.01	3.00	5.00	—
DIC	15.05	21.52	18.86	−6.47
BIC	16.58	23.83	22.72	−7.25
AIC	15.04	21.52	18.86	−6.48

Conclusion. Every information criterion favours SSEE: it fits DESI DR2 as well as ΛCDM and CPL while fixing w_0, w_a algebraically, so the parsimony of a two-parameter background is rewarded across BIC, AIC, and DIC. With the total matter density in the geometry the sound horizon matches ΛCDM and no background-structure penalty arises; the full CMB likelihood (Paper 3) independently favours SSEE at $\Delta\text{BIC} = -33.83$ (full `plik` MCMC).

B Nested Bayes Factor and Predictive Cross-Validation

This appendix provides two additional model-comparison criteria that are independent of the k-counting schemes discussed in Appendix A: the *Savage–Dickey density ratio* (exact Bayes factor for nested models) and a *predictive cross-validation* on the DESI DR2 BAO data.

B.1 Savage–Dickey density ratio

SSEE is nested within CPL: the SSEE dynamic sector is CPL evaluated at the algebraic point $(w_0, w_a) = (-0.840, -0.670)$. For nested models, the marginal Bayes factor for the null hypothesis $\mathcal{H}_0 : (w_0, w_a) = (-0.840, -0.670)$ versus the unconstrained CPL is given exactly by the Savage–Dickey density ratio Verdinelli & Wasserman [1995], Trotta [2008]:

$$B_{\rm SSEE/CPL} = \frac{p(w_0^{\rm S}, w_a^{\rm S} \mid \text{data, CPL})}{\pi(w_0^{\rm S}, w_a^{\rm S} \mid \text{CPL})} = 10.37, \qquad \ln B = +2.34. \tag{18}$$

The CPL posterior is evaluated at $(w_0^{\rm S}, w_a^{\rm S})$ using a Scott-bandwidth Gaussian KDE of $N = 500\,000$ thinned CPL chain samples (columns 3–4 of the 5-dimensional CPL chain, shape 2.5×10^6). The marginal prior on (w_0, w_a) is the truncated Gaussian $\mathcal{N}(-1, 0.5^2) \times \mathcal{N}(0, 1^2)$ clipped to the MCMC sampling box $w_0 \in (-2.5, 0.5)$, $w_a \in (-3, 2)$; truncation factors are $(0.9973, 0.9759)$, negligible corrections.

On the Jeffreys scale Trotta [2008] $\ln B = +2.34$ (> 2.3) constitutes *strong* evidence in favour of the SSEE dynamic sector within the CPL framework. The CPL posterior median ($w_0^{\rm CPL} = -0.820 \pm 0.079$, $w_a^{\rm CPL} = -0.552 \pm 0.311$) places the SSEE algebraic point at $(0.26\sigma, 0.38\sigma)$, consistent with the 0.24σ joint tension quoted in Section 6.2.

B.2 Ly-α predictive audit ($z = 2.33$)

The Ly-α point at $z = 2.33$ provides the sharpest direct test of SSEE's background expansion. At $z \sim 2$ radiation is negligible and the ratio $D_H/r_d = c\,/\,(H_0\,E(z)\,r_d)$ directly encodes the product $H_0\,E(z)\,r_d$. We evaluate it for the SSEE background and compare to the DESI DR2 measurement $D_H/r_d = 8.632 \pm 0.101$ (Table 1).

The ssee background ($\Omega_m = 0.308881$, $r_d = 147.17\,\mathbf{Mpc}$). At $z = 2.33$ the matter density sets the expansion, and the model has one: $\Omega_m = \omega_m/h^2 = 0.308881$. With $H_0^{\rm glob} = 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, (w_0, w_a) of SSEE, and the sound horizon $r_d = 147.17\,\mathrm{Mpc}$ taken directly from ω_m (CAMB; no $|w_0|$ mapping), $E(2.33) = 3.444$, $D_H(2.33) = 1281\,\mathrm{Mpc}$, and:

$$\left.\frac{D_H}{r_d}\right|_{z=2.33}^{\rm SSEE} = \frac{1281}{147.17} = 8.70 \quad (0.7\sigma \text{ from DESI}). \tag{19}$$

The full 13-point fit gives $\chi^2 = 11.41$ with the DESI covariance (11.8 diagonal), against $\chi^2 = 31.98$ (25.9) for ΛCDM at the Planck 2018 parameters, each model with its own ingredients and none refitted. The largest SSEE residual is at $z = 0.51$ (D_M/r_d pull -2.1σ); the largest ΛCDM one, at $z = 0.934$ ($+2.8\sigma$). An earlier version of this paragraph quoted $\chi^2 = 12.0$ for both models and concluded that they describe DESI DR2 equally well; those numbers had no log, and the computation above does not reproduce them.

Withdrawn scenario. Earlier versions evaluated a second "raw" scenario with the saturation $s_m = 1 + w_0 = 0.160050$ in $E(z)$ and read a large Ly-α discrepancy from it. s_m is a number of the equation of state, not a density, so that scenario is not a version of the model: it is the misplacement that produced the spurious background tensions of superseded drafts (§2.4), and it is withdrawn here.

The Savage–Dickey ratio in B.1 ($\ln B = +2.34$) then holds as a test of the *dynamical sector* under this ω_m-direct background.

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